



HRIS Project Status Report

Date: June 3, 2026 **Version:** 2.0 (Multi-Tenant Upgrade)

What is the HRIS Project? (In Simple Terms)

The HRIS (Human Resources Information System) is a portal where a company can manage its staff. Employees can clock in and out, request leaves, and view tasks, while managers and admins can approve leaves and view team attendance.

What We Replaced / Removed

Previously, the system tried to build the company structure using a complex visual flowchart builder (like drawing circles and arrows on a canvas). This was complicated, prone to errors, and hard to manage.

How we solved this:

- **What we did:** We threw away the flowchart/coordinate system entirely.
- **The new way:** We replaced it with a simple, standard spreadsheet style directory. Now, employees are simply listed in a table, and each employee has a `manager_id` pointing directly to their boss.

What Was Added (The New Features)

Software-as-a-Service (SaaS) Multi-Tenancy (Multi-Company Isolation)

Layman explanation: The system now supports multiple different companies (tenants) using the platform at the same time.

Why it matters: Each company's data is completely invisible to other companies. Two different companies can now both have an employee with ID 0001 or use similar emails, and the system will keep them isolated without mixing them up or overwriting each other.

Spreadsheet Batch Upload Tool

Layman explanation: Admins can download a template spreadsheet (.csv), fill in their employees' details (Employee ID, Name, Email, Boss ID, Hire Date), and upload it to add their entire staff in one go.

Security Smart Before-You-Upload Checker (Validator)

Layman explanation: When you choose a spreadsheet to upload, the website automatically reviews the rows before sending anything to the database. It highlights a preview grid and warns you if:

- A date is formatted incorrectly (e.g. not YYYY-MM-DD).
- An email address is typed wrong.
- There is a reporting loop (e.g. Employee A reports to Manager B, but Manager B is set to report to Employee A).

Data Engine Silent Database Updater (Upsert)

Layman explanation: When you upload the spreadsheet, the system updates matching employees with their new info or inserts new hires, without wiping out other data or logging out active users.

Compatibility Auto-Translation (Backwards Compatibility)

Layman explanation: Because we changed how employee information is structured under the hood, the older parts of the portal (like the shift clock, leave calendar, and approval pages) could have crashed. We built a translation layer that automatically translates the new spreadsheet format into the old format in the background, keeping everything working smoothly.

⚡ Summary of Recent Changes (This Session)

Session Accomplishments & Verifications

- **Added Company Separation:** Locked down the database table so records are grouped by tenant_id (the company).
- **Scopes & Security Lock:** Restricted queries and manager trees so they cannot cross-link or display across different companies.
- **Tested with Overlapping IDs:** Successfully ran a test where Company A and Company B both imported an employee with ID 0001. The database kept them perfectly isolated, ensuring they cannot see each other's profiles.
- **Built & Verified:** Compiled the frontend code to ensure the website loads fast and operates cleanly.